

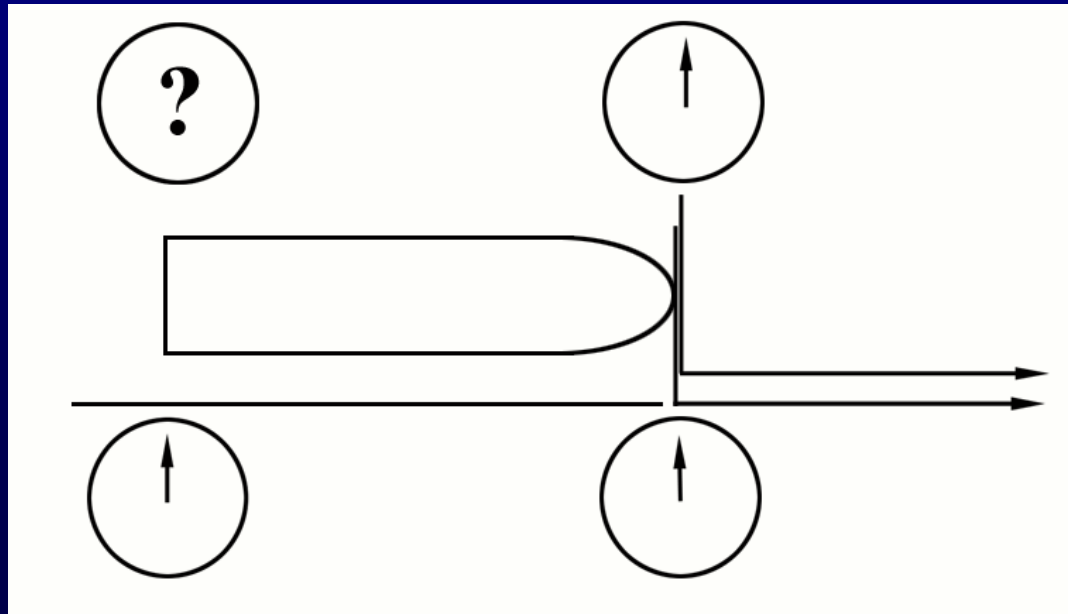
Problem 9.2

$$\ell = \frac{\ell_0}{\gamma} = \frac{10^{10} \text{ l.y.}}{10^8} = 10^2 \text{ l.y.}$$

Problem 9.3

See the supplements.

Known $\ell_0 = 200\text{m}$, $\beta = 0.60$



(a) Involved quantities: $t = 0$, $x' = -200\text{m}$, t'

$$t = \gamma \left(t' + \frac{u}{c^2} x' \right) \quad t' = -\frac{u}{c^2} x' \\ = \frac{0.6 \times 200\text{m}}{c} = 0.40\mu\text{s}$$

(b) (the tail of the ship to reach us)

$$x = 0, x' = -200\text{m}, t$$

$$x' = \gamma (x - ut) \quad t = -\frac{x'}{\gamma u} = 0.89\mu\text{s}$$

Or directly $T = \frac{\ell}{u} = \frac{\ell_0}{\gamma u}$

(c) (the tail of the ship is besides us)

$$x = 0, x' = -200\text{m}, t'$$

$$x = \gamma(x' + ut') \quad t' = \frac{-x'}{u} = 1.1\mu\text{s}$$

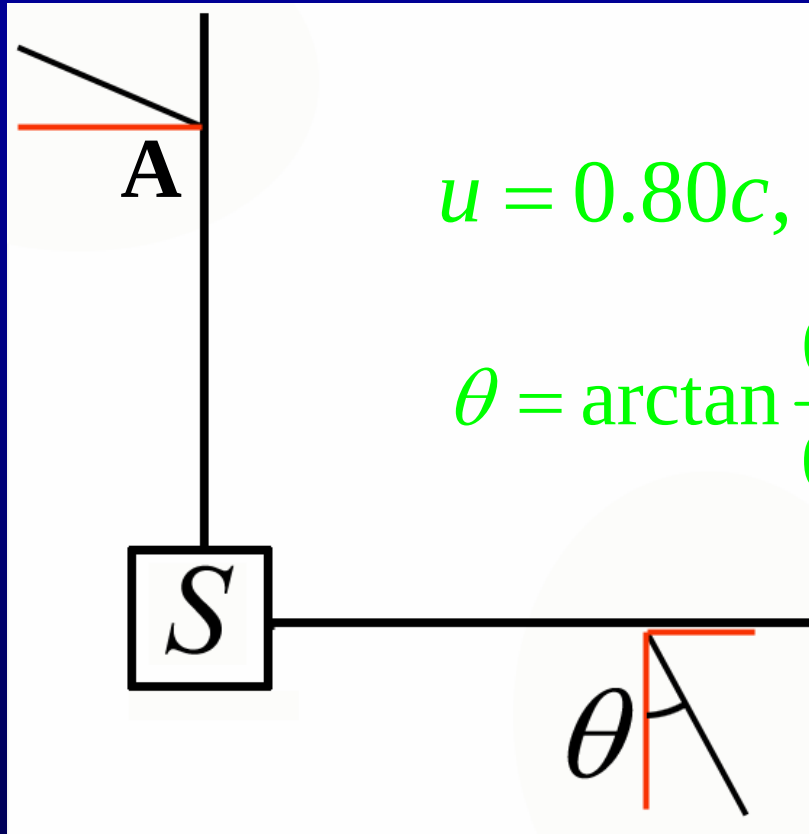
Problem 9.6

$$\beta = \frac{\left(\frac{\lambda}{\lambda_0}\right)^2 - 1}{\left(\frac{\lambda}{\lambda_0}\right)^2 + 1} = 0.742, \quad v = 2.23 \times 10^8 \text{ m/s}$$

Problem 9.9

$$\frac{\lambda_o}{\lambda_e} = 3 = \sqrt{\frac{1 + \beta}{1 - \beta}}; \quad \beta = \frac{3^2 - 1}{3^2 + 1} = 0.8$$

Problem 9.13



A as S'.

$$u = 0.80c, v'_x = -u, v'_y = \frac{v_y}{\gamma} = 0.48c$$

$$\theta = \arctan \frac{0.48}{0.80} = 31^\circ$$

v_{BA} is at symmetric position as shown in the figure.

Problem 9.16 Conservation of mass-energy is

$$m_{n0}c^2 = m_{p0}c^2 + m_e c^2 + E_v$$

or $E_e + E_v = m_{n0}c^2 - m_{p0}c^2 \equiv \Delta \quad (*)$

Because of $\mathbf{p}_e + \mathbf{p}_v = \mathbf{0}$, $E_v = cp_v$, we have

$$\frac{1}{c} \sqrt{E_e^2 - E_0^2} = \frac{1}{c} E_v \quad \leftarrow E_e^2 = p_e^2 c^2 + E_0^2$$

That is $E_e^2 - E_v^2 = E_0^2 \quad (**)$

$$E_e - E_v = E_0^2 / \Delta \quad (***)$$

From (*) and (*), we have**

$$E_e = \frac{1}{2} \left(\Delta + \frac{E_0^2}{\Delta} \right)$$

$$K = E_e - E_0 = \frac{1}{2} \left(\Delta + \frac{E_0^2}{\Delta} \right) - E_0$$

$$= \frac{1}{2\Delta} (\Delta - E_0)^2$$

$$= \frac{1}{2(939.565 - 938.272)}$$

$$\times (939.565 - 938.272 - 0.511)^2 \text{ MeV}$$

$$= 0.236 \text{ MeV}$$

***Problem 9.17** $E_{k\Sigma} = 250\text{MeV}$

$$E_{k\pi} = 57.4\text{MeV}$$

$$E_{kn} = 60.7\text{MeV}$$

$$\theta = 9.7^\circ$$

Problem 9.19

$$p^2 c^2 = E^2 - m_0^2 c^4 = E^2 \left[1 - \frac{m_0^2}{m^2} \right] = E^2 \left[1 - \frac{1}{\gamma^2} \right]$$

$$p = \frac{E}{c} \left(1 - \frac{1}{\gamma^2} \right)^{1/2} \approx \frac{E}{c} \left(1 - \frac{1}{2\gamma^2} \right)$$

$$(c) \beta = \sqrt{1 - \frac{1}{\gamma^2}} \approx 1 - \frac{1}{2\gamma^2} = 0.99995;$$

$$p = mv = \frac{mc^2}{c} \beta = 10500 \times 0.99995 \\ = 10499.475 \frac{\text{MeV}}{c};$$

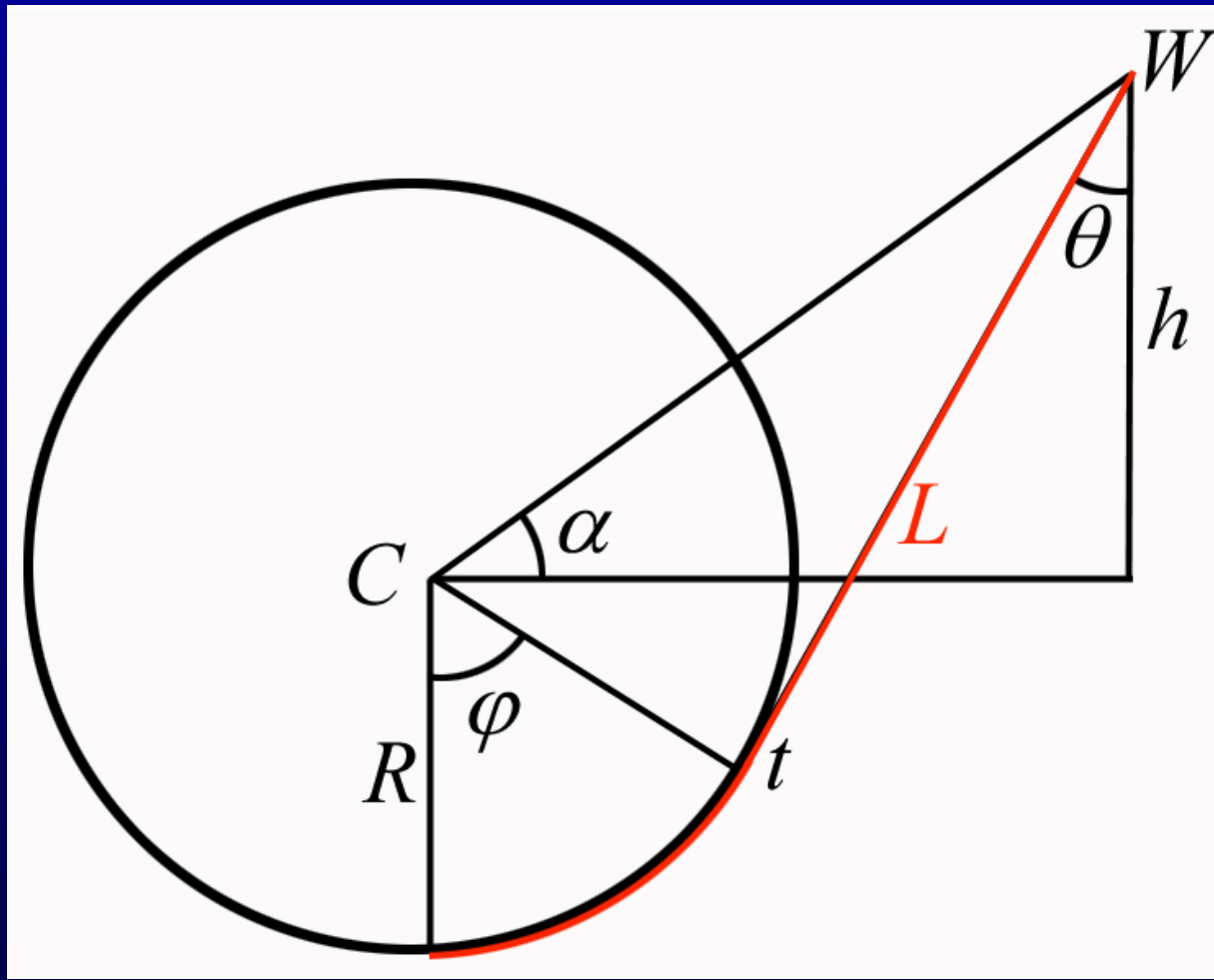
$$(d) x' = 0, t = \gamma t' = 2 \times 10^{-4} \text{ s}$$

$$(e) s = vt = \beta ct = 0.99995 \times 3 \times 10^8 \times 2 \times 10^{-4} \text{ m} \\ = 5.9997 \times 10^4 \text{ m}$$

problem 10.1

(a) -40 ; (b) 575 ; (c) N/A

Problem 10.4



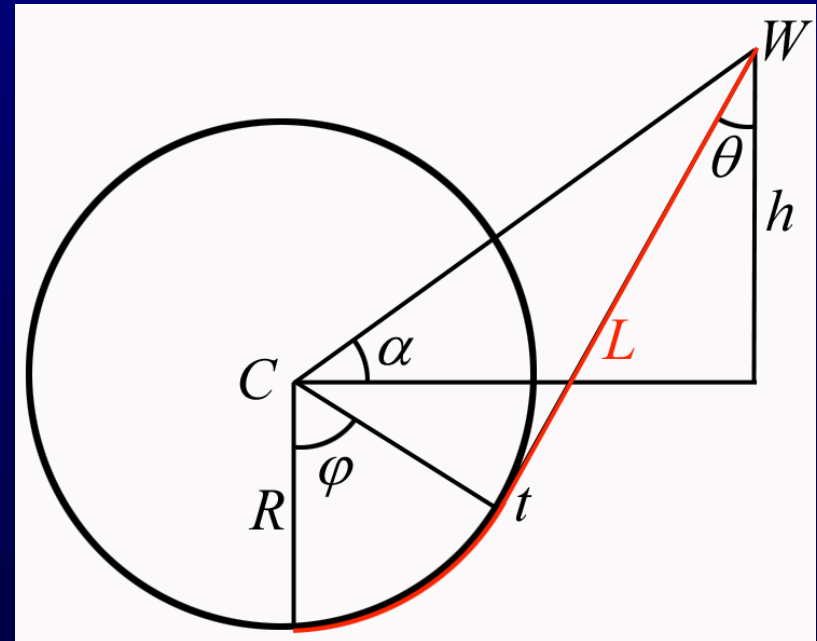
Neglect the weight of the belt. $\overline{tW}$ is straight.

The axis C of the cylinder not to be moved by thermal expansions or contractions of the cylinder and belt. $\overline{CW}$ and h are constant.

$$h = (L - R\varphi - R \tan \theta) \cos \theta$$

1. misleading $d\theta = 0$

$$\frac{dL}{dT} - \frac{dR}{dT} \left(\frac{\pi}{2} - \theta + \tan \theta \right) = 0$$



**Using the definition of expansion coefficient,
we have**

$$\frac{dL}{dT} = L\alpha_{\text{steel}} \qquad \frac{dR}{dT} = R\alpha_{\text{Al}}$$

$$L\alpha_{\text{steel}} - R\alpha_{\text{Al}} \left(\frac{\pi}{2} - \theta + \tan \theta \right) = 0$$

$$R = \frac{2L}{\pi - 2\theta + 2 \tan \theta} \frac{\alpha_{\text{steel}}}{\alpha_{\text{Al}}} \approx 0.63\text{m}$$

as indicated by Halliday

2. Exact treatment

In $\triangle CtW$,

$$R = \overline{CW} \cos(\alpha + \theta)$$

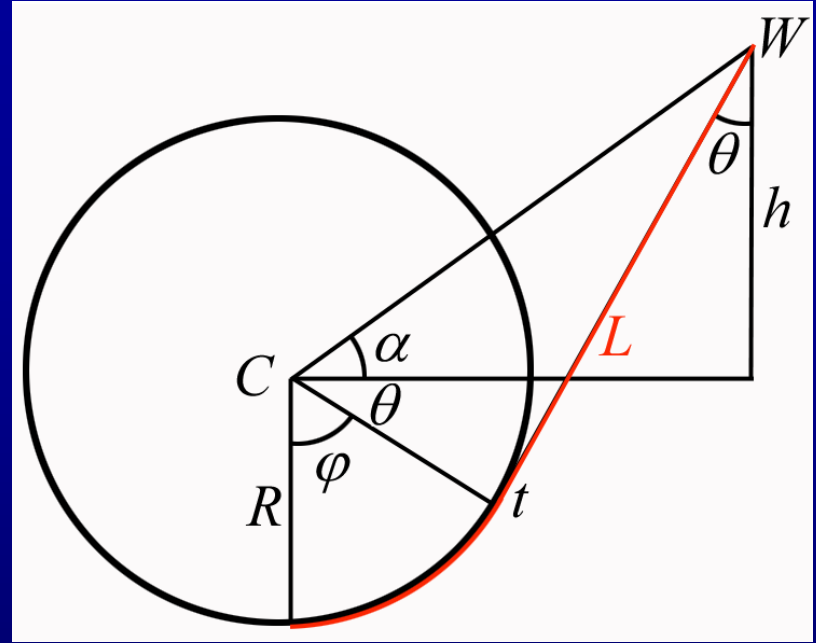
$$dR = -\overline{CW} \sin(\alpha + \theta) d\theta$$

$$= -tW d\theta$$

$$= -(L - R\varphi) d\theta$$

$$\frac{d\theta}{dT} = -\frac{1}{(L - R\varphi)} \frac{dR}{dT}$$

$$\frac{d\varphi}{dT} = \frac{d}{dT} \left(\frac{\pi}{2} - \theta \right)$$



$$\frac{dh}{dT} = \left(\frac{dL}{dT} - \frac{dR}{dT} \varphi - R \frac{d\varphi}{dT} - \frac{dR}{dT} \tan \theta - R \sec^2 \theta \frac{d\theta}{dT} \right) \cos \theta$$

$$- \left(L - R\varphi - R \tan \theta \right) \sin \theta \frac{d\theta}{dT} \longrightarrow 0$$

$$\frac{d\theta}{dT} = - \frac{1}{(L - R\varphi)} \frac{dR}{dT} \quad \frac{d\varphi}{dT} = \frac{d}{dT} \left(\frac{\pi}{2} - \theta \right)$$

$$\frac{dL}{dT} = \frac{dR}{dT} \left[\frac{\pi}{2} - \theta + \frac{R}{L - R\varphi} + \tan \theta - \frac{R}{L - R\varphi} \sec^2 \theta \right]$$

$$- \left(L - R\varphi - R \tan \theta \right) \tan \theta \frac{1}{L - R\varphi}$$

$$\alpha_{\text{steel}} L = \alpha_{\text{Al}} R \left(\frac{\pi}{2} - \theta \right)$$

$$R = \frac{L}{\frac{\pi}{2} - \theta} \frac{\alpha_{\text{steel}}}{\alpha_{\text{Al}}}$$

$$= 1.7\text{m}$$

~~$$R = \frac{L}{\frac{\pi}{2} - \theta + \tan \theta} \frac{\alpha_{\text{steel}}}{\alpha_{\text{Al}}} \approx 0.63\text{m}$$~~

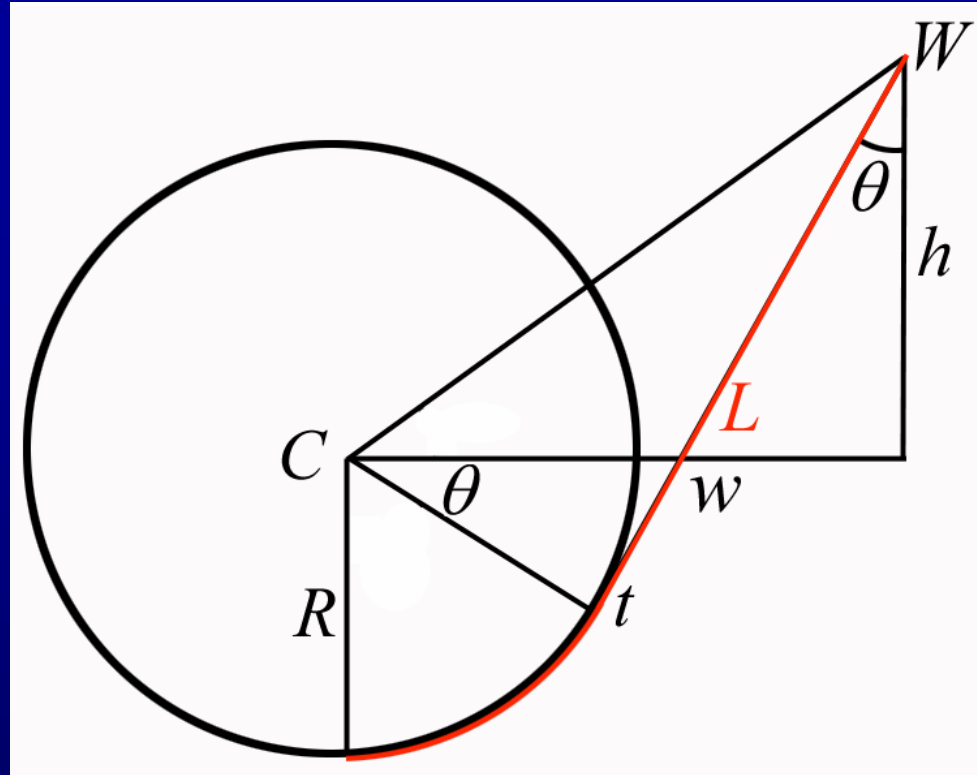
Problem 10.4

$$\overline{tW} \equiv \ell_0$$

$$L = \ell_0 + \left(\frac{\pi}{2} - \theta \right) R$$

$$h = \ell_0 \cos \theta - R \sin \theta$$

$$w = \ell_0 \sin \theta + R \cos \theta$$



(Dec 19 2003 6c101)

$$R = w \cos \theta - h \sin \theta$$

$$\ell_0 = w \sin \theta + h \cos \theta$$

$$\left(\frac{dR}{d\theta} = -\ell_0 \right)$$

$$\frac{d\ell_0}{d\theta} = R \qquad \frac{d\ell_0}{dT} = \frac{d\ell_0}{d\theta} \frac{d\theta}{dT} = R \frac{d\theta}{dT}$$

$$R\alpha_{\text{Al}} = \frac{dR}{dT}$$

$$L\alpha_{\text{steel}} = \frac{dL}{dT}$$

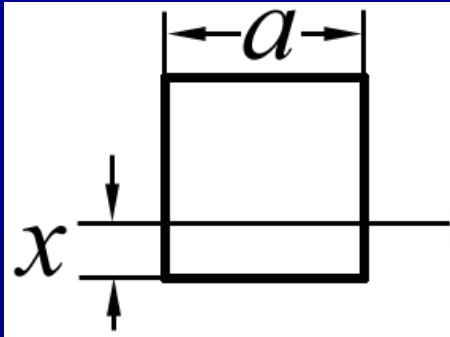
$$= \frac{d\left[\ell_0 + \left(\frac{\pi}{2} - \theta\right)R\right]}{dT}$$

$$= R\frac{d\theta}{dT} + \left(\frac{\pi}{2} - \theta\right)\frac{dR}{dT} - \frac{d\theta}{dT}R$$

$$= \left(\frac{\pi}{2} - \theta\right)\frac{dR}{dT}$$

$$R = \frac{L}{\frac{\pi}{2} - \theta} \frac{\alpha_{\text{steel}}}{\alpha_{\text{Al}}}$$

problem 10.5



The depth submerged x is determined by

$$a^2 x \rho_{\text{Hg}} g = a^3 \rho_{\text{Al}} g$$

$$x = a \frac{\rho_{\text{Al}}}{\rho_{\text{Hg}}} = 20\text{cm} \frac{2.70}{13.6} = 3.97\text{cm}$$

Meanwhile

$$\ln x = \ln a + \ln \rho_{\text{Al}} - \ln \rho_{\text{Hg}}$$

$$\frac{d \ln x}{dT} = \frac{1}{a} \frac{da}{dT} + \frac{1}{\rho_{\text{Al}}} \frac{d\rho_{\text{Al}}}{dT} - \frac{1}{\rho_{\text{Hg}}} \frac{d\rho_{\text{Hg}}}{dT}$$

$$\begin{aligned}
\frac{1}{\rho_{\text{Al}}} \frac{d\rho_{\text{Al}}}{dT} &= \frac{1}{\frac{\cancel{m}}{V}} \frac{d}{dT} \left(\frac{\cancel{m}}{V} \right) \\
&= V \left(-\frac{1}{V^2} \right) \frac{dV}{dT} \\
&= -\frac{1}{V} \frac{dV}{dT} \\
&= -\beta
\end{aligned}$$

$$\int d \ln x = \int dT (\alpha_{\text{Al}} - \beta_{\text{Al}} + \beta_{\text{Hg}})$$

$$= (\alpha_{\text{Al}} - 3\alpha_{\text{Al}} + \beta_{\text{Hg}}) \int dT$$

$$\ln \frac{x}{x_0} = (\alpha_{\text{Al}} - 3\alpha_{\text{Al}} + \beta_{\text{Hg}}) \Delta T$$

$$= (-2 \times 2.3 \times 10^{-5} + 1.8 \times 10^{-4}) \times 50$$

$$= 6.7 \times 10^{-3}$$

$$\Delta x = x_0 (e^{6.7 \times 10^{-3}} - 1)$$

$$= x_0 \times 6.7 \times 10^{-3} = 0.27 \text{ mm}$$

problem 11.1 (a) No (external heat),

(b) Yes(work), (elastic collision requires a full rebound.)

(c) Yes, $\Delta U_{\text{in}} + mg\Delta h = 0$,

$$\frac{\Delta U}{m} = -\Delta(gh) = 9.8 \times 9.5 \text{ J/kg} = 93 \text{ J/kg}$$

$$(d) \quad \Delta T = \frac{\Delta U}{c} = \frac{93 \text{ J/kg}}{0.50 \text{ kJ/(kg} \cdot \text{K)}} = 0.19 \text{ K}$$

*** problem 11.2**

$$p = AT^3/V$$

$$U = BT^n \ln(V/V_0) + f(T)$$

***Using the equation in p213**

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p$$

we have

$$\frac{BT^n}{V} = \frac{T \cdot 3AT^2}{V} - \frac{AT^3}{V}$$

which leads to $n = 3$, $B = 2A$.

Problem 11.5 In equilibrium

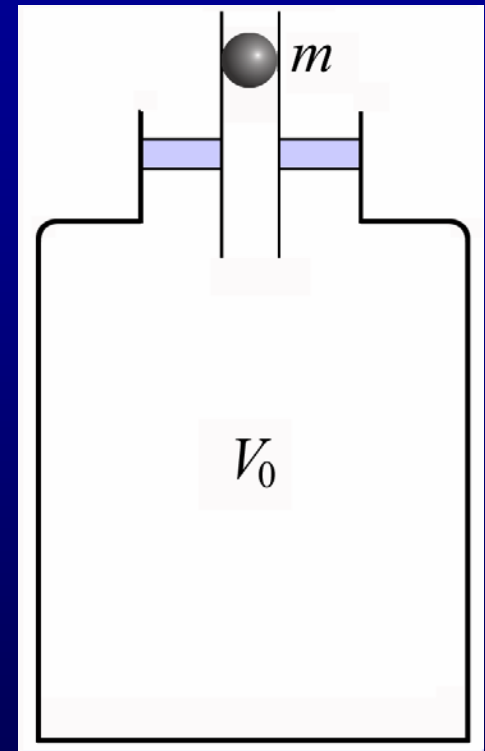
$$p_1 = p_0 + \frac{mg}{A}$$

$$pV^\gamma = \text{const.}$$

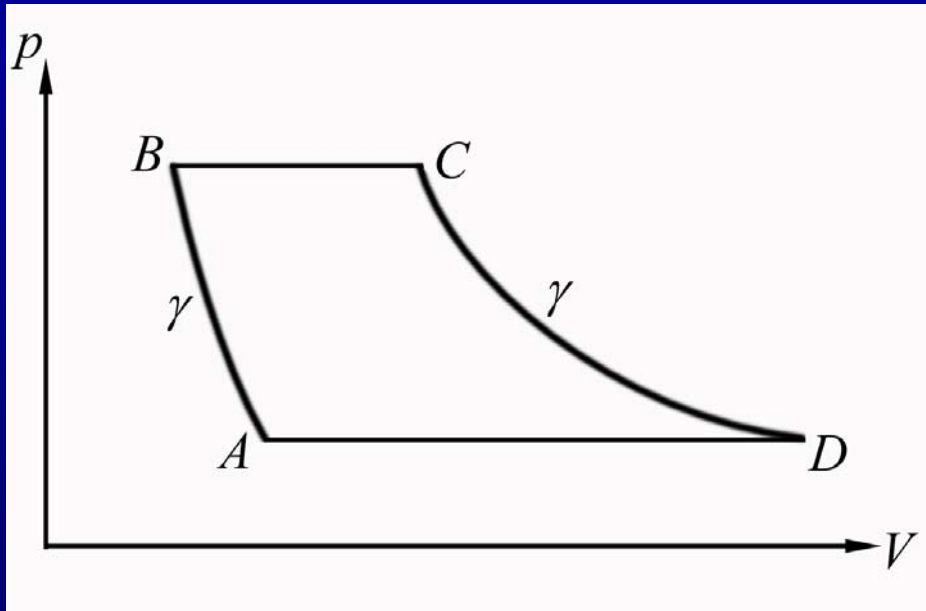
$$\frac{dp}{p} = -\gamma \frac{dV}{V} = -\gamma \frac{Ax}{V_0}$$

$$f = dp \cdot A = -\gamma \frac{pA^2}{V_0} x$$

$$\omega^2 = \frac{\gamma A^2 p}{mV_0} \quad \nu = \frac{\omega}{2\pi}$$



problem 11.9 Joule cycle



$$\Delta Q_{BC} = C_p (T_C - T_B)$$

$$\Delta Q_{AD} = C_p (T_D - T_A)$$

$$\eta = 1 - \frac{T_D - T_A}{T_C - T_B}$$

Along adiabatics

$$pV^\gamma = C \quad p\left(\frac{T}{p}\right)^\gamma = C' \quad Tp^{\frac{1}{\gamma}-1} = C''$$

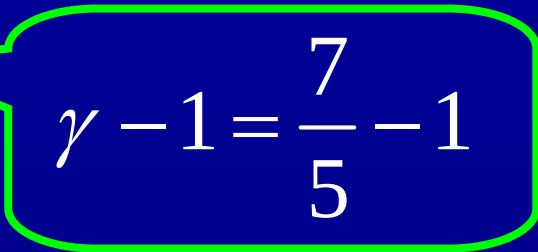
$$T_B p_B^{\frac{1-\gamma}{\gamma}} = T_A p_A^{\frac{1-\gamma}{\gamma}} ; \quad T_C p_C^{\frac{1-\gamma}{\gamma}} = T_D p_D^{\frac{1-\gamma}{\gamma}}$$

$$\frac{T_B}{T_C} = \frac{T_A}{T_D} ; \quad \frac{T_D}{T_C} = \left(\frac{p_B}{p_A} \right)^{\frac{1-\gamma}{\gamma}}$$

So

$$\eta = 1 - \left(\frac{p_A}{p_B} \right)^{\frac{\gamma-1}{\gamma}}$$

problem 11.10

$$\eta = 1 - \left(\frac{V_1}{V_2} \right)^{2/5}$$

$$\gamma - 1 = \frac{7}{5} - 1$$